

Psychophysiological Effects of Guided Imagery Based Intervention on the Academic Development of Children: An Open-Source Heart Rate Variability Analysis Toolbox

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Abstract—Several studies have assessed the efficacy of psychotherapies aimed at enhancing children’s psychological well-being. However, few have used psychophysiological parameters, e.g. Heart Rate Variability (HRV), to elucidate their underlying mechanisms. Traditional HRV analysis involves time-consuming hand-driven correction of artefacts in noisy electrocardiogram (ECG) recordings. Thus, this study proposes an automated HRV analysis pipeline to extract HRV parameters. The pipeline includes a pre-processing using an adaptive ACC filtering and noise detection, R-peak detection and a time-series based automatic artefact correction. Four state-of-the-art R-peak detectors were benchmarked, with Pan-Tompkins++ R-peak detector showing higher performance (100% sensitivity for sitting and maths test Glasgow University datasets). The time-series automatic artefact correction detected normal beats with 99% of accuracy and classified low temporally displaced misaligned R-peaks with 100% of accuracy. This pipeline was validated against Kubios HRV Premium 4.1.0, extracting 8 out of 12 acceptable HRV metrics. This open-source toolbox significantly reduced time-consuming hand-driven artefact correction for large ECG recordings.

Index Terms—Heart Rate Variability Analysis, Psychophysiological Indicators, Open-Source.

I. INTRODUCTION

In 2019, the World Health Organization reported that approximately 301 million individuals were affected by anxiety disorders, a prevalent mental health condition¹. Yet, only about 1 in 4 of these individuals received a treatment for these conditions. Mental imagery interventions have proven to be a cost-effective and efficient response to mitigate this mental health crisis. They contribute to promoting well-being and hold promise as a treatment for psychopathology [1]. Mental imagery is defined as the experience of perception without sensory input, which involves the reciprocal activation of physiological, behavioral, and emotional responses. This concept resonates with the ideas presented by Kuppens et al. [2], who state “that emotions are both biologically grounded and culturally shaped”. Research has consistently shown that the intensity and perspective of images encountered in mental imagery can elicit stronger emotional reactions in adults [3]. These findings have prompted discussions about the potential benefits and challenges of applying such interventions to younger age groups. Moreover, there is a limited number of studies that have provided evidence on the impact of guided imagery on physiological indicators, as a way of understanding the psychophysiological mechanisms involved in emotional regulation and well-being in children.

Mental imagery is characterized by an intrinsic and dynamic *mind-body connection*. This connection is enabled by a complex psychophysiological network, as evidenced by observable physiological changes [4]. Notably, the Central Autonomic Network (CAN) is a functional unit within the Central Nervous System (CNS) that seems to regulate goal-directed behaviors and emotional adaptability [5]. In terms of functionality, this network, which has the capacity to modulate emotions, is deeply connected with the ability to adaptively modify perceptual and emotional brain activities in response to evolving environmental demands. The CAN’s main output is channeled through the preganglionic sympathetic and parasympathetic neurons, collectively known as the Autonomic Nervous System (ANS) [5]. Sympathetic neurons are responsible for the release of norepinephrine, which increases heart rate (HR) in the so-called fight-or-flight response under stress. In contrast, acetylcholine released from vagal nerve endings has the opposite effect. The interplay of these ANS outputs from the CAN at the sino-atrial node of the heart results in complex beat-to-beat variability [6]. This is also known as Heart Rate Variability (HRV), which allows for the observation of ANS activity using HRV analysis. In fact, HRV derives from the fluctuations in the time intervals between adjacent heartbeats. This information can be extracted from ECG signals acquired from individuals using electrodes and ECG sensors. In turn, the ECG reflects the electrical activity of the heart, where each R-peak of the QRS complex indicates a heartbeat. Evidence suggests that cardiac vagal tone is associated with adaptive responsiveness to the environment [4]. Moreover, individuals who hold greater emotional regulation tend to exhibit higher levels of vagally-mediated HRV [6].

Within this framework, the RegularMente project [1] is being developed as a randomized controlled trial (RCT) that evaluates the effects of guided imagery intervention on the psychophysiological parameters of primary school students, including HRV. While obtaining HRV from ECG typically requires only R-peak detection to compute RR intervals (RRi), the RegularMente dataset needs additional processing steps. This dataset comprises ECG recordings of children in a classroom setting, acquired using wearable devices. Such ecological and unrestrained setting may introduce artefacts in the wearables data. In particular, motion artefacts and noise due to technical issues can interfere with R-peak detection, compromising the result’s accuracy.

Hence, the goal of this work is to develop a pipeline for HRV analysis, which comprises: 1) R-peak detection; 2) Auto-

¹“Anxiety disorders — who.int”: <https://www.who.int/news-room/fact-sheets/detail/anxiety-disorders> [Accessed 12-05-2024].

matic RRi time series artefact correction; 3) Noise detection; and 4) noise removal based on accelerometer (ACC) data in order to overcome obstacles found in RegularMente. To validate our HRV analysis tool, we compare its results with ECG dataset containing annotated QRS complexes similar to the RegularMente clinical trial dataset. Subsequently, we apply the signal processing pipeline to the RegularMente dataset and extract relevant psychophysiological metrics.

II. STATE-OF-THE-ART

HRV has been demonstrated to be informative of the impact of multiple psychotherapies, particularly within the adult pathological demographic [6]. Despite this, there remains a lack of evidence supporting its use in child populations. Regardless of the study population, to assess the effectiveness of such therapies, psychophysiological studies underscore the necessity of establishing dependable biosignal markers that can elucidate the physiological mechanisms driving emotional responses. Giannakakis et al. [7] have identified numerous time-domain characteristics and spectral analysis metrics that are suitable for the characterization of some psychophysiological responses. Among the time-domain features that mirror RR variability, both the Standard Deviation of NN intervals (SDNN) and the Root Mean Square of Successive Differences (RMSSD) — an index influenced by vagal modulation — are significantly diminished under stress. Moreover, spectral analysis revealed a significant reduction in the high-frequency (HF) spectrum of HRV during stress, mirroring a decrease in vagal tone. The association of individual physiological indicators with particular time-domain or spectral analysis parameters has been substantiated across extensive clinical settings. Malik et al. [8] have illustrated a robust correlation between time and frequency domain metrics within a 24-hour HRV analysis, substantiating their relevance in characterizing physiological markers for a range of clinical scenarios, such as cardiovascular diseases. Nonetheless, the precise extraction of each feature requires rigorous preprocessing and HRV signal extraction procedures.

Mathersul et al. [9] conducted a RCT to evaluate the efficacy of yoga-based interventions for individuals with posttraumatic stress disorder. They utilized self-reported measures of emotion regulation alongside HRV analysis to monitor the effects of this trauma-focused psychological therapy on US Veterans. The ambulatory cardiac data was thoroughly analyzed, and relevant features were extracted using Kubios HRV Premium 4.1.0 [10]. This benchmark software for HRV analysis and feature extraction facilitated the conclusion that HRV may serve as an indicator of autonomic resilience and the ability to adapt to stressors among these US Veterans.

The growing interest in using HRV analysis to describe several psychophysiological mechanisms has catalyzed the development of numerous software tools, both commercial and open-source. Nevertheless, even though there is an extensive selection of HRV analysis tools available, such as Kubios HRV [10] and SinusCor [11], Lu et al. [12] highlighted

in their comprehensive review the persisting ambiguities associated with data collection, the computational models of HRV analysis software, and physiological variables that may result in diminished signal quality, thereby impacting the integrity of HRV analysis. Despite their unique features, all of them require a signal pre-processing and R-peak detection.

A. Pre-processing

Indeed, ECG signals are often contaminated by various types of noise. This noise includes both low-frequency (LF) components, e.g. respiration, and HF components, e.g. interference from power lines. Therefore, a band-pass filter is typically employed as the initial pre-processing step. The objective of a band-pass filter is to eliminate both the low and HF noise that does not contribute to the detection of QRS complexes, while preserving the frequencies that contain relevant information that is essential for the subsequent processing steps. Both the Pan-Tompkins [13] and Hamilton [14] R-peak detectors use a similar pre-processing step that includes a band-pass filter with a pass-band of [5; 15] Hz. Elgendi et al. [15] conducted a study to determine the most effective band-pass filter for R-peak detection. They examined seven different frequency bands proposed in literature and concluded that the optimal passing band for detecting R-peaks is [8; 20] Hz.

Although band-pass filters are able to remove LF and HF noise, motion artefact may not be removed by band-pass filtering, as it shares similar frequencies with the QRS complex. Consequently, motion artefact removal becomes an essential part in signal pre-processing. Martinek et al. [16] proposed the use of an adaptive filter, specifically the Least Mean Square (LMS) algorithm, to filter out motion artefacts. The LMS algorithm seeks to adjust the filter based on both input and output to minimize the mean square error [17]. To understand whether currently available softwares include a feature for motion artefact removal, we have reviewed five mainstream HRV analysis software into our state-of-the-art, namely Kubios HRV (by Kubios Oy), AcqKnowledge® (by BIOPAC), HRV Add-on (by PLUX), LabChart8 (by ADInstruments) and the SinusCor HRV analysis toolbox [11]. However, none of these software integrate motion artefact removal.

Previously mentioned pre-processing steps are only effective if the signal quality is acceptable. Conversely, in the case of wearable or remote monitoring ECG devices, there can be instances where the quality of ECG recordings are extremely low (e.g. when the electrode loses contact with the patient), leading to a segment of the signal that cannot be interpreted [18]. This type of ECG segments should be excluded for further analysis to prevent inaccuracy of the psychophysiological parameters. For this purpose, Rahman et. al [18] have tested various Statistical Signal Quality Index (SSQI) as a method for noise detection, i.e. differentiating noisy ECG segment from the clean ones. They conclude that when compared with manual binary annotation, where ECGs were classified as noisy or noise-free, the accuracy of SSQI varies depending on the dataset and the type of SSQI used. Specifically, the highest accuracy for MIT/BIH arrhythmia (MITDB), Tele

ECG, and CINC 2014 dataset was obtained using Signal-to-Noise Ratio SQI (SQI_{snr}) (97.51%), SQI_{snr} (73.40%) and Higher-Order Statistics-SQI (SQI_{hos}) (77.20%), respectively [18]. Effectively, none of the five softwares mentioned earlier include SQI-based noise detection feature. Although Kubios HRV employs an RRI-based noise detection algorithm, the implementation details of this algorithm remain unpublished.

B. R-Peak Detection

The detection of R-peaks is a critical component in HRV analysis for accurate feature extraction. The R-peak detection algorithm that is most prevalently adopted was proposed by Pan and Tompkins [13]. This algorithm incorporates an adaptive multi-thresholding approach for R-peak detection, which relies on the digital evaluation of the slope, width, and amplitude of the QRS complexes. Additionally, the algorithm includes a preprocessing phase that uses a bandpass filter to diminish signal interference, as well as a dual-threshold and search-back method for identifying missed R-peaks. This method adaptively adjusts based on the morphology of the QRS complex and HR. When tested against the standard 24-hour MITDB [19], the algorithm demonstrated a sensitivity (Se) exceeding 99.30%. Nonetheless, Porr and Howell [20] identified certain limitations in the original Pan and Tompkins R-peak detection algorithm. Their analysis revealed that the Se for the MITDB annotated by the PhysioBank [19] annotation codes, when compared to the claimed values, was 12.85%. Over the years, numerous other R-peak detection algorithms have been proposed, each varying in complexity.

The SinusCor HRV analysis toolbox [11] employs a straightforward derivative-based method for detecting R-peaks using a fixed threshold. This basic algorithm for R-peak detection may result in numerous inaccuracies due to motion artefacts and the nonstationarity behaviour of the ECG signal. To address the shortcomings of fixed thresholding, Kew et al. [21] developed a simple variable threshold method and assessed its performance utilizing the MITDB long-term ECG dataset [19], across five subjects. The fixed threshold method achieved an average accuracy of 90.1%, whereas the variable threshold method reached 98.8% accuracy for those specific datasets. Nonetheless, the majority of R-peak detection algorithms struggle to maintain high accuracy within wearable ECG datasets, owing to the heterogeneity and noise artefacts they contain. Therefore, from a pool of freely available state-of-the-art R-peak detectors, further studies should be conducted to determine which R-peak detector is most suitable for managing large, noisy ECG recordings, such as those found in the RegularMente dataset.

C. Automatic RRI Time Series Artefact Correction

Upon obtaining R-peaks time series from R-peak detectors, RRI can be calculated. In fact, artefacts in RRI can occur due to either the patient's physiological condition or the data acquisition equipment itself. It is crucial to first identify and then correct these artefacts to minimize inaccuracy on HRV parameters. Common artefacts include extra,

missed or misaligned beats as well as ectopic beats [22]. Traditionally, an experienced cardiologist is responsible for artefact identification and manual correction through visual inspection. This process is particularly time-consuming and requires specialized training, therefore, an automated RRI time series artefact correction process is highly beneficial. Some softwares provides automatic artefact identification but not correction, e.g. AcqKnowledge® by BIOPAC can detect and mark significant differences between successive R-peaks, indicating possible artefact locations. Later, the user needs to correct the marked artefact manually. Other softwares seem to only detect and correct certain types of artefacts, e.g. LabChart8 by AD Instruments is only able to detect and remove ectopic beats but not other types of artefacts. In contrast, among all five explored softwares, Kubios HRV provides the most complete automatic RRI time series artefact correction algorithm, which encompasses both the identification and correction of all types of artefacts mentioned above [22]. The identification of artefacts is based on time-varying thresholds estimated from distribution of successive RRI differences combined with a novel beat classification scheme, which allows identification of extra, missed or misaligned beat and ectopic beats. According to the type of detected artefact, automatic RRI time series artefact correction will be performed. The implementation details of this automatic RRI time series artefact correction algorithm were published by Lipponen et al. [22], making it readily available for incorporation when necessary.

III. METHODS AND MATERIALS

The primary drawback of wearable devices is compromised signal quality due to common sources of noise such as motion artefacts, data loss, and electrical interference. These issues significantly hinder feature extraction, thereby affecting the reliability of HRV parameters. HRV analysis requires accurate time intervals between successive normal-to-normal RRI. However, the RR time series often contains artefacts, i.e. abnormal RRI values, particularly during extended HRV measurements outside laboratory settings. Consequently, this often leads to extra, missed, or misaligned beat detections, as well as atrial and ventricular ectopic beats, disrupting RR time series analysis. Manually inspecting and correcting each ECG segment for these noise sources to obtain reliable HRV parameters can be extremely time-consuming and complex. Therefore, we incorporated additional processing steps into our HRV analysis pipeline (Figure 1) instead of relying solely on a conventional R-peak detector. Each processing step was subsequently validated to ensure the extraction of relevant HRV parameters from the RegularMente dataset.

A. Pre-processing

1) *Adaptive ACC Filtering*: During real-time ECG measurements using wearables, motion artefacts caused by electrode contact significantly impact signal quality and feature extraction. To address this issue, we propose integrating a Block Least Mean Square (BLMS) algorithm [23] into our HRV analysis workflow. Using a three-axis ACC on the arm,

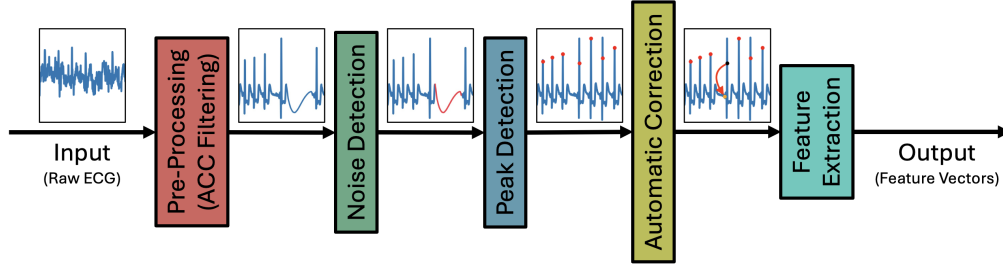


Fig. 1. HRV analysis pipeline including signal processing and feature extraction presented in this work.

the BLMS algorithm employs a coefficient vector that is updated based on the block gradient estimate at each occurrence ($\mu = 0.46$). Huang et al. [23] introduced the concept that, instead of recalculating filter coefficients for each sample as in the traditional LMS algorithm, the computational complexity is reduced by estimating the weight coefficients in each block (`block_size` = 2). This approach provides a dynamic ECG signal motion artefact's filtering process.

2) *Noise Detection*: For large real-time ECG recordings contaminated with many types of noise, feature extraction can be challenging. Recently, researchers proposed using SSQIs to improve accuracy and reliability of HRV analysis (Section II-A). The most common SSQI are the skewness, SQI_{skew} [18]:

$$SQI_{skew} = \frac{1}{N} \sum_{i=1}^N \left[\frac{(x_i - \bar{x})}{\sigma} \right]^3 \quad (1)$$

and the peakedness of kurtosis, which is a statistical measure of the distribution of the inspected data throughout the mean, SQI_{kur} [18]:

$$SQI_{kur} = \frac{1}{N} \sum_{i=1}^N \left[\frac{(x_i - \bar{x})}{\sigma} \right]^4 \quad (2)$$

where x denotes the ECG signal with N sample points, $\bar{x}$ and σ represent the mean value and standard deviation of signal x , respectively. Nardelli et al. [24] decided to combine both SSQIs metrics, SQI_{kur} and SQI_{skew} , resulting in a novel SSQI, SQI_{hos} [24]:

$$SQI_{hos} = |SQI_{skew}| \times \frac{SQI_{kur}}{5} \quad (3)$$

Indeed, SQI_{hos} and SQI_{kur} demonstrate good generalization capability in identifying noise and noise-free segments across various ECG recordings [18]. However, for HF noise, the entire ECG signal is classified as noise-free, compromising the specificity of these SSQIs. Motion artefacts and other HF noises introduce a high degree of variability in the distribution of RRi, resulting in reduced performance of R-peak detection and feature extraction. Therefore, we suggest using an SSQI dependent on the variation of RRi, $cSQI$; to enhance the accuracy of HRV analysis [25]:

$$cSQI = \frac{\hat{\sigma}_{RR}}{\hat{\mu}_{RR}} \quad (4)$$

where $\hat{\mu}_{RR}$ and $\hat{\sigma}_{RR}$ are the empirical estimates of the mean and standard deviation of the distribution of the RRi in a ECG segment, respectively. This leads to a novel method by incorporating $cSQI$ and SQI_{hos} to our HRV analysis pipeline. It is important to note that Kubios HRV Premium 4.1.0 does not include this noise detection technique [10]. The identification criteria of SSQI are given as, where the threshold was determined empirically [18] [25]:

$$ECG \begin{cases} \text{optimal}, & cSQI \geq 0.45, SQI_{hos} \leq 0.8 \\ \text{unqualified}, & \text{otherwise} \end{cases} \quad (5)$$

B. State-of-the-Art R-peak Detectors

Our R-peak detection algorithm selection showcases several popular heartbeat detectors, each employing diverse strategies to correctly identify all R-peaks and extract relevant HRV parameters.

1) *Pan and Tompkins*: The conventional R-peak detection algorithm proposed by Pan and Tompkins [13] begins with pre-processing using a bandpass filter within a range of [5; 15] Hz and implemented as a first order Butterworth IIR filter. Next, a differentiator is used to highlight the sharp slopes corresponding to the QRS complex followed by a squaring process. After, a moving window average with 150 ms window is applied to obtain waveform feature information in addition to the slope of the R-peak. Peaks at least 300 ms apart are identified in the pre-processed signal and classified as either noise or a QRS complex based on an adaptive threshold. To ensure peak detection, two thresholds ($Threshold_1$ and $Threshold_2$) are used, with a search-back approach implemented if no QRS complex is identified within a specified window. Subsequently, during the search-back process, the average RRi is computed using the last nine R-peaks.

2) *Hamilton*: Based on Pan and Tompkins R-peak detection algorithm, Hamilton [14] developed a similar R-peak detector but with some variations. Firstly, the bandpass filter has a bandpass of [8; 16] Hz and a moving window average with 80 ms window. The pre-processing step relies on a rectification step instead of a squaring process and uses a narrow moving average window. For R-peak detection, if a peak occurs within 360 ms of a previous detection and has a maximum slope less than half of the maximum slope of the previous detection, it is assumed to be a T-wave. Secondly, if an RRi equal to 1.5 times the average RRi has elapsed since the most recent

detection, within that RRI there was a peak that was larger than half the detection threshold, and the peak followed the preceding detection by at least 360 ms, classify that peak as a R-peak. The detection threshold is measured using the mean of the average QRS peak (*SPK*) and the average noise peak (*NPK*) buffers.

3) *Pan-Tompkins++*: Imtiaz et al. [26] developed an R-peak detection algorithm aimed to improve the Pan and Tompkins algorithm by addressing its limitations. Their pre-processing stage includes a bandpass filter set of [5; 18] Hz, followed by an N-point moving average filter applied to the squared signal before using a 60 ms moving average window.

For R-peak detection, an extra threshold ($Threshold_3$) is employed alongside $Threshold_1$ and $Threshold_2$ to minimize false negative (FN) peaks by searching back within a pre-defined RRI interval to detect any missing QRS complexes. If the maximal point in that window is greater than $Threshold_3$, it is classified as an R-peak and the *SPK* and *NPK* are adjusted. The detection rules for *SPK* and *NPK* are the same as Pan and Tompkins algorithm [13]. The lower threshold was slightly lowered by Imtiaz et al. to detect low amplitude R-peaks that the Pan and Tompkins algorithm seem to skip due to the high value for the lower threshold during searching. Furthermore, several false positive (FP) R-peaks that are close to each other seem to be detected by the Pan and Tompkins algorithm. Thus, using a RRI check that measures the mean slope over the preceding 70 ms window of the current peak and the most recent detected R-peaks reduced the amount of FP beats after a real R-peak is detected. If the slope within the current waveform is less than 60% of that of the preceding QRS waveform, it is classified as a T-wave; otherwise, it is a R-peak. Finally, the Pan and Tompkins algorithm struggles to detect R-peaks when exceptionally high peaks disrupt the regular pattern. Therefore, after locating the window 360 ms following the preceding R-peak detection, an R-peak is identified as the highest point in that window that exceeds 20% of the $Threshold_2$. Upon detection of a R-peak, the *SPK* and *NPK* values are automatically adjusted according to predefined detection rules. Using the MITDB, Imtiaz et al. achieved a Se of 99.47% compared to 99.17% achieved by the original Pan and Tompkins algorithm. Other datasets were tested and showed that Pan-Tompkins++ outperforms the conventional R-peak detector presented by Pan and Tompkins and also the R-peak detection algorithm presented by Hamilton [14].

4) *NeuroKit*: The NeuroKit R-peak detector [27] operates differently from the abovementioned algorithms by utilizing a high-pass filtered raw ECG signal with a low-cut frequency of 0.5 Hz. Firstly, it computes the gradient of the ECG signal and establishes a gradient threshold to detect QRS complexes using a 100 ms smoothing window. Secondly, it identifies R-peaks within the QRS complexes by detecting local maxima without employing a search process. It is important to note that the algorithm includes a minimum delay within each QRS complex to prevent FP R-peaks, similar to the approach implemented in the Pan-Tompkins++ detection algorithm.

C. Automatic RRI Time Series Artefact Correction

HRV parameters are notably affected by HRV artefacts, including extra, misaligned, and missing beats. Therefore, an automatic method for artefact detection and correction is essential. Lipponen et al. [22] introduced an algorithm designed specifically for this purpose. The detection of artefacts relies on two variable thresholds. The first threshold distinguishes between artefacts and normal beats, with adaptability achieved through the time-varying distribution of the differences between successive RR intervals (dRRs). Using dRRs, ectopic beats and single missing, short, or long beats are detected. For the second threshold, as extra beats and detector errors are not easily discriminated using dRRs, the difference between individual RR intervals (mRRs) is employed, ensuring adaptability as well. Additionally, Lipponen et al. [22] present a decision algorithm to classify each type of erroneous beat. After identifying erroneous beats, the invalid RRI times are corrected, relying on the classification of each type of erroneous beat. Finally, the effect of simulated artefacts and artefact correction on time and frequency domain feature extraction was evaluated. Indeed, several HRV analysis toolboxes focus solely on artefact detection without providing correction mechanisms. Kubios HRV offers automatic RRI time series artefact correction, but this feature is only available in the Premium version. Therefore, implementing this functionality in our open-source pipeline will allow other researchers to freely utilize it in practical scenarios, such as with the RegularMente dataset.

D. Evaluation Criteria

1) *Pre-processing*: For the pre-processing evaluation, we applied a statistical systematization to compare the time and frequency domain features extracted by our HRV analysis pipeline with those from Kubios HRV Premium 4.1.0, the gold standard in HRV analysis. We categorized the results into three distinct groups: *Optimal*, *Acceptable*, and *Divergent*.

The mean values for our HRV analysis pipeline ($\overline{P_x}$) and for Kubios HRV ($\overline{K_x}$) and their standard deviations; $\sigma(P_x)$ and $\sigma(K_x)$, respectively, were computed where x corresponds to a specific HRV parameter. Features will be extracted from the RegularMente dataset for each subject, focusing on the use of adaptive filtering with ACC and noise detection (Section III-A). The statistical classification follows the methodology described by Gomes et al. [28], using the $\overline{P_x}$ and $\overline{K_x}$ difference of HRV parameters, Δ_x ; to verify the similarity between the two software tools:

TABLE I
STATISTICAL TAXONOMY OF Δ_x BETWEEN OUR HRV ANALYSIS PIPELINE AND KUBIOS HRV.

$\Delta_x = 0$	$0 < \Delta_x \leq \sigma(K_x)$	$ \Delta_x > \sigma(K_x)$
<i>Divergent</i>	<i>Acceptable</i>	<i>Optimal</i>

2) *R-peak Detection*: Four distinct state-of-the-art R-peak detectors (Section III-B) were evaluated using the annotated Glasgow University Database (GUDb) [20]. By comparing

the detected R-peaks to the annotations of the GUDB, we obtain the number of correctly detected R-peaks, true positive (TP); the number of wrong labeled R-peaks given by each R-peak detector algorithm, FP; and the number of R-peaks that are not detected by the algorithm, FN. The performance of the detector algorithms was compared using the number of correctly detected R-peaks to the total number of known R-peaks; Se [29]:

$$Se = \frac{TP}{TP + FN} \quad (6)$$

The ratio between number of correctly detected R-peaks to the total number of detected R-peaks; Positive Predictivity (PPr) [29]:

$$PPr = \frac{TP}{TP + FP} \quad (7)$$

The ratio between the total number of incorrectly R-peaks to the total number of known R-peaks; Detection Error Rate (DER) [29]:

$$DER = \frac{FN + FP}{TP + FN} \quad (8)$$

Finally, as proposed by Zhao et al [29], the root mean square time difference between the n -th location of the correctly detected R-peak, D_n ; and the nearest annotated R-peak, K_n ; Annotated-Detected Error (ADE) [29]:

$$ADE = \sqrt{\frac{1}{TP} \sum_{n=1}^{TP} (K_n - D_n)^2} \quad (9)$$

ADE reaches zero if and only if the positions of all R-peak locations in the correctly detected QRS complexes perfectly match the annotated peak locations.

3) *Automatic RRi Time Series Artefact Correction*: To assess the performance of the implemented automatic RRi time series artefact correction algorithm [22], artefacts will be artificially generated. Missed beats were assessed by removing detected beats at indexes $k = \alpha n$, where $n = \{1, 2, 3, \dots\}$, while extra beats will be generated after the index k with a integer time-step $\alpha = 15$. For misaligned beats, Lipponen et al. [22] proposed using temporal displacements (Δt) at index k , given by $\Delta t = q \times \text{RMSSD}$, where q corresponds to a specific displacement level for $q = \{2, 4, 8\}$. The simulated artefacts will be assessed using the GUDB to evaluate the automatic RRi time series artefact correction algorithm. The detection rate will be determined by the percentage of artifacts that have been successfully detected. Meanwhile, the classification accuracy will be measured by the percentage of these detected artifacts that were correctly categorized into their specific types.

IV. RESULTS

A. Data Description

Our HRV analysis aims to extract time and frequency domain features from the RegularMente dataset comprised

of 240 participants around 8 - 11 years old with a total of 1000 records. Data was collected using the BITalino device by PLUX [30] (Figure 3); the ECG signals were recorded with an ECG sensors attached to the upper chest with two pre-gelled Ag/AgCl electrodes and the ACC data was recorded using a comfortable arm band, placed in the non dominant arm, above the elbow. Each single-lead ECG recording has a duration of 40 minutes with data collected using a sampling rate of 100 Hz and a resolution of 10-bit. All the recordings were provided in h format (.h file) and text format (.csv file) from OpenSignals software [30]. Due to the lack of annotations in the RegularMente dataset, it was necessary to incorporate additional datasets.

To ensure the validity of each stage within the HRV analysis pipeline, supplementary datasets bearing resemblance to those characteristics exhibited by the RegularMente dataset were employed. For the evaluation of R-peak detection and automatic RRi time series artefact correction algorithm, the GUDB has 25 participants around 18 years old with a total of 125 records. This dataset consists of 2 minutes ECG recordings each performing five different tasks (sitting, using a tablet to perform a maths test, walking on a treadmill, using a hand-bike and jogging) was used [20]. The wireless acquisition board used to create GUDB had a sampling rate of 250 Hz and a resolution of 24-bit. The chest strap with V2-V1 and Einthoven II leads were chosen. High-precision beat location annotations were assigned to all ECG recordings, apart from those excessively affected by noise interference.

B. State-of-the-Art R-Peak Detectors Evaluation

The results of the comparison of four R-peak detection algorithms, as mentioned in Section III-D2, are presented in Table II. The data consists of 105 ECG from GUDB, with 20244 beats in total.

In terms of detected number of beats, the general tendency is that all the R-peak detectors underestimate beats when compared to Pan-Tompkins++, which seems to detect all R-peaks for all GUDBs except for jogging (Table II). In Figure 2, an example of R-peak detection for a 20 seconds ECG from the Maths Test is presented. It demonstrates that all algorithms, except Pan-Tompkins++, failed to detect the first R-peak. This failure is likely due to the adaptive nature of these R-peak detectors, which adjust their parameters based on the characteristics of the incoming ECG signal. Notably, Pan-Tompkins++ has a searchback feature which could mitigate this problem. The reduced performance during jogging for Pan-Tompkins++ could be attributed to several factors, such as increased HR and motion artefacts. Interestingly, the original Pan and Tompkins algorithm had the lowest difference of beats compared to the annotated in jogging (Pan and Tompkins: +30, Table II). This fact might due to a characteristic with Pan and Tompkins algorithm already referred previously in Section III-B3, where this algorithm detects several FP R-peaks that are close to each other. Indeed, the ECG signal in jogging can be contaminated by motion artefact, which makes Pan and Tompkins algorithm to mistakenly identify several

TABLE II
EVALUATION OF FOUR STATE-OF-THE-ART R-PEAK DETECTION ALGORITHMS USING GUDB.

GUDB Dataset	Method	Annotated Number of Beats	Number of Beats	Difference of Beats	Se (%)	PPr (%)	DER (%)	ADE (ms)
Sitting	Hamilton	3362	3333	-29	99.17	100	0.01	2.37
	NeuroKit		3354	-8	99.75	99.96	0	0
	Pan and Tompkins		3297	-65	16.2	16.4	1.66	5.45
	Pan-Tompkins++		3459	0	100	100	0	1.22
Maths Test	Hamilton	3725	3713	-12	99.54	99.86	0.01	2.1
	NeuroKit		3718	-7	99.79	99.97	0	0
	Pan and Tompkins		3689	-36	21	21.14	1.57	6.45
	Pan-Tompkins++		3727	0	100	100	0	0.99
Walking	Hamilton	3834	3813	-21	99.44	99.97	0.01	2.33
	NeuroKit		3682	-152	95.85	99.85	0.04	0
	Pan and Tompkins		3818	-16	29.47	29.53	1.41	8.43
	Pan-Tompkins++		3882	-2	99.92	99.97	0	1.12
Hand-Bike	Hamilton	3731	3685	-46	98.32	99.53	0.02	2.16
	NeuroKit		3758	27	98.64	97.9	0.04	0
	Pan and Tompkins		3866	135	35.96	34.64	1.33	5.01
	Pan-Tompkins++		3841	-1	99.84	99.87	0	1.11
Jogging	Hamilton	5592	5229	-363	88.89	94.36	0.16	1.98
	NeuroKit		5365	-227	94.17	97.33	0.08	0
	Pan and Tompkins		5622	30	39.87	39.68	1.21	8.33
	Pan-Tompkins++		5561	-175	94.96	97.63	0.07	0.88

motion artefacts as R-peaks. Meanwhile, we observe that Pan-Tompkins++ demonstrated greater performance in detecting R-peaks, closely matching the annotated ones.

To evaluate the accuracy of our R-peak detectors, the Se was used (Equation 6). Hamilton and NeuroKit showed Se consistently above 95% across all GUDB activities except for jogging (Hamilton: 88.89%, NeuroKit: 94.17%). Pan and Tompkins has significantly lower Se, especially for less intense activities such as sitting and maths test. However, it is surprising to observe that the Se of Pan and Tompkins algorithm actually increases with exercise intensity (sitting: 16.2%, jogging: 39.87%). This contradicts the expectation that Se would decrease as exercise intensity rises, given the increased data loss and motion artefacts resulting from the instability of ECG sensors. Pan-Tompkins++ shows higher overall Se in all GUDB activities, despite its reduced Se for jogging (Pan-Tompkins++: 94.96%) compared to the other activities. Pan-Tompkins++ appears to be an enhanced version of the Pan and Tompkins R-peak detector. In addition to incorporating a bandpass filter followed by an N-point moving average filter to eliminate noise without discarding significant signal components, it utilizes $Threshold_1$, $Threshold_2$, and $Threshold_3$ to distinguish between R-peaks. This likely explains its high Se across all activities in the GUDB.

To assess the precision of each state-of-the-art R-peak detectors we selected the PPr (Equation 7) and DER (Equation 8) metrics. Firstly, Hamilton and Pan-Tompkins++ consistently showed high performance across all GUDB activities, maintaining a PPr approximately to 100%. NeuroKit also shows high PPr values (sitting: 99.96 %, maths test: 99.97%), despite its conservative approach to detect R-peak based on local maxima. Similar to the trend observed in the Se values (Table II), the PPr decreases for higher intensity activities in the GUDB. This highlights the effectiveness of adaptive

threshold-based R-peak detectors, such as Pan-Tompkins++ and Hamilton. This reason could corroborate the null values of DER for Pan-Tompkins++ (except for jogging: 0.07%) and low values of DER for Hamilton R-peak detectors. However, the Pan and Tompkins algorithm exhibits the highest DER values overall. This might be due to its tendency to detect many FP R-peaks that are close to each other (Equation 8) and an increased number of FN R-peaks, as it struggles to detect R-peaks after long periods when there are no significant changes in signal pattern.

Lastly, ADE measures the temporal distance between detected and annotated R-peaks (Equation 9). NeuroKit consistently shows a null ADE across all GUDB activities, which might indicate high accuracy in R-peak detection contrary to Hamilton's ADE which ranges from 1.98 ms to 2.37 ms. Pan-Tompkins++ has a low ADE values ranging from 0.88 ms to 1.22 ms. As expected, due to its limitations, Pan and Tompkins R-peak detector has the highest values of ADE. Several factors might have been contributed to these results, such as the discrete nature of digital signals and the tendency of Pan and Tompkins algorithm to detect the S-wave instead of the R-peak, as illustrated in Figure 2. The Pan and Tompkins algorithm also falsely detects the T-wave, as noted by Imtiaz et. al [26]. Overall, NeuroKit achieves the lowest ADE, indicating its strong performance. However, the automatic R-peak detector used for annotated beats seems to follow the same principles as the NeuroKit R-peak detector, which might explain its null ADE values. This is further evidenced by its reduced performance in other evaluation metrics (Table 2).

C. Automatic RRI Time Series Artefact Correction Evaluation

The validation results for the automatic RRI time series artefact correction algorithm developed by Lipponen et. al. [22] is shown in Table III, which illustrates the detection and

TABLE IV
EVALUATION RESULTS FOR THE HRV PARAMETERS EXTRACTED WITH ADAPTIVE ACC FILTERING AND NOISE DETECTION.

Time and Frequency Domain Evaluation Metrics with Adaptive ACC Filtering and Noise Detection				
Parameter	Unit	Our Work	Kubios HRV	$ \Delta_x $
Mean HR	[bpm]	115.11 $\pm$ 11.01	109.52 $\pm$ 10.90	5.59
Minimum HR	[bpm]	56.50 $\pm$ 4.76	34.92 $\pm$ 6.83	21.58
Maximum HR	[bpm]	226.29 $\pm$ 30.67	240.00 $\pm$ 0.00	13.71
Mean NNI	[ms]	587.10 $\pm$ 67.62	551.94 $\pm$ 51.58	35.15
RMSSD	[ms]	114.84 $\pm$ 67.13	111.45 $\pm$ 38.44	3.39
pNN50	[%]	34.91 $\pm$ 24.05	31.02 $\pm$ 13.37	3.89
SDNN	[ms]	97.49 $\pm$ 47.74	117.99 $\pm$ 40.09	20.50
Low Frequency Power (LF)	[ms ²]	4994.50 $\pm$ 3950.79	9224.40 $\pm$ 9061.34	4229.90
High Frequency Power (HF)	[ms ²]	20869.35 $\pm$ 31618.16	6902.37 $\pm$ 6433.32	13966.98
Total Power	[ms ²]	5895.60 $\pm$ 5452.71	17241.92 $\pm$ 16225.69	11346.32
LF/HF Ratio	[–]	1.37 $\pm$ 1.49	1.45 $\pm$ 0.56	0.08
TINN	[ms]	171.16 $\pm$ 66.40	1349.60 $\pm$ 273.89	1178.44

classification rate of simulated misaligned beats for multiple displacements Δt , alongside missed and extra beats. The data comprises 63 ECG from GUDB including sitting, maths test and walking activities to simulate the classroom setting found in RegularMente dataset. In total, there are 10921 normal beats and 700 simulated abnormal beats for each artefact type. Detection and classification accuracy gives the percentage of beats which have been detected as artefacts and classified into the correct class respectively.

The detection and classification rate exceed 98% for normal, missed and extra beats, except for classification rate of 95.32% of missed beats which could be more difficult.

When it comes to misaligned beats, the classification and detection rate of the algorithm change as Δt changes (Section III-D3). The results (Table III) suggest that, for the misaligned beats, as the level of misalignment increases, i.e. higher q values, the algorithm’s detection rate improves from 62.95% to 99.36%, but its classification rate decreases from 100% to 89.06%, suggesting that misaligned beats with factor $q = 8$ are easier to be detect but harder to be classified correctly.

TABLE III
PERFORMANCE OF AUTOMATIC RRI TIME SERIES ARTEFACT CORRECTION ALGORITHM TO DETECT AND CLASSIFY NORMAL BEATS AND SIMULATED ARTEFACTS.

Beat Types	Number of beats	Mean Δt (ms)	Classification (%)	Detection (%)
Normal	10921	-	-	99.00
Missed	700	-	95.32	98.41
Extra	700	-	98.57	98.27
Misaligned ($q = 2$)	700	59.83	100.00	62.95
Misaligned ($q = 4$)	700	120.05	97.51	98.24
Misaligned ($q = 8$)	700	240.52	89.06	99.36

D. HRV Parameters Extraction

The results of HRV metrics extraction in this study are benchmarked against the gold-standard Kubios HRV Premium 4.1.0. To determine whether the adaptive ACC filtering or noise detection algorithm has a greater impact on the extraction of HRV parameters, we conducted three HRV metrics extractions on the same population for different pipeline combinations (Figure 1): only ACC filtering, only SSQI-based

noise detection (Equation 5), and both adaptive ACC filtering and SSQI-based noise detection. Our hypothesis is that the combined pre-processing steps yield the best results.

The data comprises five subjects from the RegularMente dataset, each with a duration of approximately 1h30m. The results revealed that when compared to Kubios HRV, the HRV features extracted 8 *Divergent* metrics with ACC filtering alone (Table V), 6 *Divergent* metrics with SSQI-based noise detection alone (Table VI), and 4 *Divergent* metrics with a combination of ACC filtering and noise detection (Table IV). These findings suggest that the combination of the algorithms is able to produce results that are most closely aligned with Kubios HRV.

V. DISCUSSION

This study was conducted to address some limitations in extracting HRV parameters from large real-time ECG recordings collected using wearable devices, such as those in the RegularMente dataset. As shown in Table IV, our HRV analysis pipeline shows minor discrepancies in the HRV parameters compared to those obtained using the benchmark HRV analysis software, Kubios HRV Premium 4.1.0. Unlike Kubios HRV and other commercial HRV analysis tools, our code will be freely available, making it accessible for anyone to use in their HRV analyses to extract relevant features. Details of our open-source code will become available in BioSPPy [31].

The Python implementation of our code offers several advantages, including its open-source nature and the flexibility to modify it according to specific requirements. The current pipeline is capable of efficiently handling a substantial volume of data, thereby achieving a batch processing effect. Still, despite these benefits, the user has to interact through the Python program through a command-line interface, which might hinder its user experience. To address this limitation, future work could focus on the development of a more intuitive Graphical User Interface (GUI), particularly for data selection and analysis. For instance, users can easily analyze numerous ECG recordings by simply dragging and dropping a folder containing the recordings into the GUI. This proposed

improvement aligns with our prior goal of facilitating batch processing for the broader scientific community. This approach would make the toolbox more accessible and user-friendly, thereby promoting its widespread adoption.

Motion artefacts, data loss, and other technical issues commonly contaminate biosignals recorded with wearable devices. This problem is particularly pronounced in the RegularMente dataset, which mainly consists of recordings from children in a school environment. For the adaptive ACC filtering, each parameter was adjusted empirically, which may limit its applicability to other datasets. Parameter optimization could involve an iterative approach to minimize the differences in a selected HRV parameter shown in Table IV. Zhao et al. [29] demonstrated that this process is insensitive to the initially defined values and step lengths for each parameter of the implemented BLMS algorithm. Furthermore, to enhance the performance of adaptive ACC filtering, the small-step assumption and stability conditions should be applied to simplify the parameter selection process [17]. The same limitation applies to the SSQI-based noise detection step, as SSQIs often require empirical threshold settings to distinguish between noisy and noise-free ECG segments, which may become ineffective when used with different datasets (Equation 5). Several machine-learning-based SQIs (MLSQIs), such as support vector machines (SVM) and linear discriminant analysis (LDA), have been proposed to overcome this issue, achieving high accuracy and specificity [18]. While MLSQIs are effective for detecting noise in ECG segments, machine-learning models require extensive labeled datasets, which are not widely available.

Among the state-of-the-art R-peak detectors evaluated, Pan-Tompkins++ exhibits higher Se under various conditions in the GUDB, achieving a Se of 100% for the sitting and maths test activities (Table III), suggesting it detects all true R-peaks. This warrants further analysis since the algorithm typically employs high temporal tolerances exceeding 100 ms, potentially masking the true Se value. For Hamilton and Neurokit R-peak detectors, it was observed lower Se values compared to Pan-Tompkins++ R-peak detector (Table 2) for sitting and maths test activities, which might be explained by the impact of baseline wandering. This phenomenon occurs due to the rotation of the heart's electrical axis during the respiratory cycle, which alters the thoracic impedance distribution [12]. Alternatively, NeuroKit R-peak detector demonstrates higher precision, consistently showing a null ADE across all GUDBs. It is important to note that the GUDB annotations were initially generated by an automatic detection algorithm that identified all local maxima through simple comparison of neighboring values. Subsequently, these annotations were visually inspected and corrected. This methodology aligns with the rationale employed in the NeuroKit R-peak detector algorithm, which may explain the observed ADE values (Section III-B). Moreover, researchers have shown that Pan-Tompkins++ reveals reduced execution time compared to conventional R-peak detectors, making it more suitable for real-time cardiac monitoring applications [26].

Furthermore, the performance of nearly all evaluated R-peak detectors deteriorates as the intensity of exercise escalates, with the most severe decline observed during the most intense activity, e.g. jogging. This could be attributed to several factors, e.g. motion artefacts. To improve the adaptability to different activities, Belkadi et al. [32] suggest a deep neural network approach for R-peak detection, which involves an automatic detector that does not require neither preprocessing nor human intervention. This innovative algorithm is based on a semisupervised stacked autoencoder deep neural network, which has been proven to outperform traditional R-peak detection algorithms in terms of accuracy. By employing deep learning, this algorithm extracts features directly from unprocessed data, eliminating the need for expert intervention. The algorithm exhibited a Se of 99.66% across 1,470,284 beats [32], as tested on six publicly accessible benchmark datasets [19], potentially improving results for high-intensity exercises.

Our findings on the automatic RRI time series artefact correction algorithm (Table III) deviated from Lipponen et al. [22]. While they observed an increase in classification accuracy of misaligned beats with increasing Δt , we found the opposite pattern. This discrepancy can be attributed to the substantial number of artefacts generated with $k = 15n$ (approximately 6% of total beats) during our assessment instead of $k = 100n$ (1% of total beats) used by Lipponen et al. (Section III-D3). The increased amount of artefacts can influence the dRRs and mRRs used in the algorithm to calculate classification thresholds, leading to the incorrect classification of misaligned peaks ($q = 8$) as missed peaks.

When adaptive ACC filtering and SSQI-based noise detection are combined, 4 *Divergent* metrics were identified after HRV parameters extraction (Table IV). These metrics include the Minimum and Maximum HR, High Frequency Power (HF), and Triangular Interpolation of NN interval histogram (TINN). There appears to be a discrepancy between our work and Kubios HRV in terms of the Minimum and Maximum HR values. Our Minimum HR value is closer to physiological interval than that of Kubios HRV (Our Work: 56.50 bpm, Kubios HRV: 34.92 bpm), which could suggest that Kubios HRV might have missed some beats (FN), leading to higher RR intervals and consequently, a lower HR. In terms of Maximum HR, both our work and Kubios HRV detected an exceptionally high HR (Our Work: 226.29 bpm, Kubios HRV: 240.00 bpm). These two HRV parameters are indeed sensitive to outliers as they represent the extreme values of HR during the measurement period. In fact, any artefacts or noise that persist through each 1 hour and 30 minutes ECG recording, despite the pre-processing and automatic RRI time series artefact correction algorithm, may explain the non-physiological Minimum and Maximum HR values observed.

For HF Power, our result is significantly higher than Kubios HRV (Our Work: 20869.35 ms^2 , Kubios HRV: 6902.37 ms^2). Previous research has established a strong correlation between HF power and the proportion of NN50, i.e. the number of times successive heartbeat intervals exceed 50 ms divided by the total number of RRI (pNN50), as well as RMSSD [33].

Hence, we would anticipate *Divergent* results for pNN50 and RMSSD, since HF power is *Divergent*. However, when comparing our results with those of Kubios HRV, the pNN50 and RMSSD values are within *Acceptable* ranges, contradicting the expected correlation. This suggests a potential discrepancy in the relationship between HF power and these time-domain measures, demanding further investigation. Besides, even two artefacts within a 5-minute segment can significantly distort TINN value [33]. Hence, artefacts or noises present in these 1h30m ECG recordings could result in a broader or narrower distribution of NN intervals, thereby influencing the TINN.

VI. CONCLUSION

With this work, we aim to develop an HRV analysis toolbox to address limitations in processing large real-world ECG recordings, such as the impact of motion artefacts and other typical noise sources, thereby extracting relevant physiological indicators. Our HRV analysis pipeline includes a pre-processing step with adaptive ACC filtering and SSQI-based noise detection, features not included in the benchmark Kubios HRV Premium 4.1.0 toolbox. Additionally, the Pan-Tompkins++ R-peak detector was selected for its higher performance compared to other state-of-the-art R-peak detectors, achieving a Se of 100% in the GUDBs for activities such as sitting and performing a maths test. We also incorporated an automatic RRI time series artefact correction to accurately detect and classify R-peaks, achieving an outstanding 100% classification for misaligned beats ($q = 2$) and 99.36% detection for misaligned beats ($q = 8$). This step reduces the time-consuming and complex hand-driven correction of artefacts in large and noisy ECG recordings outside laboratory settings. Therefore, we present a robust and automatic HRV analysis pipeline that can extract 8 out of 12 acceptable HRV parameters compared to Kubios HRV Premium 4.1.0 from the RegularMente dataset. Further improvements, such as machine learning-based approaches to define empirically defined thresholds, could help achieve the same or even greater performance as the gold-standard HRV analysis toolbox. The code will soon become freely available in BioSPPy [31].

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VII. APPENDIX

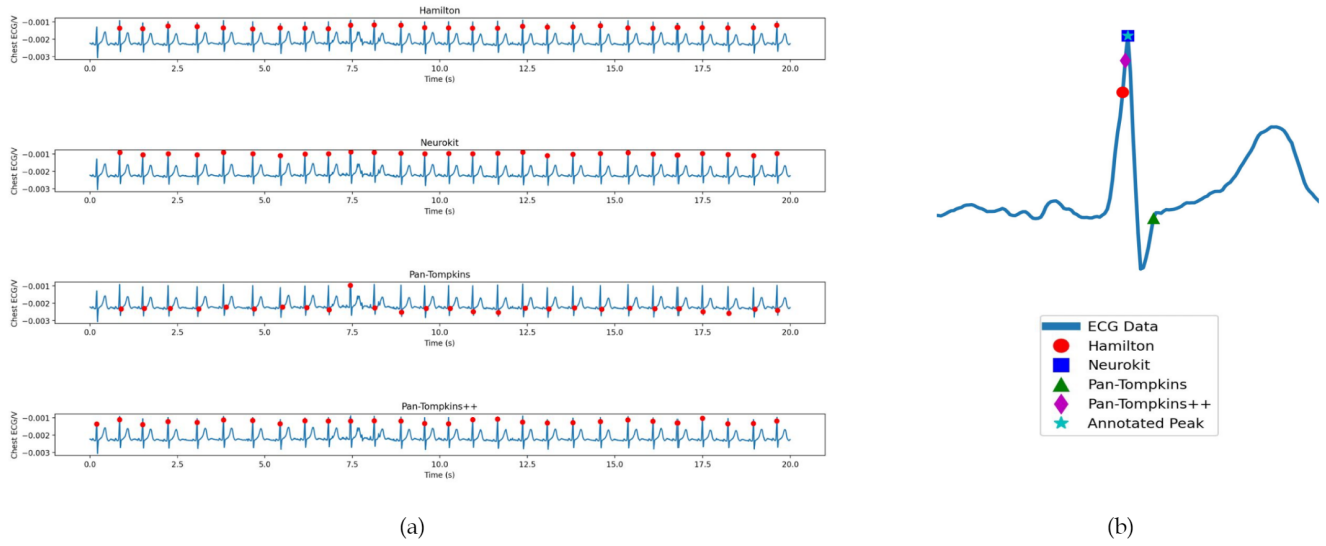


Fig. 2. Performance of state-of-the-art R-peak detectors (a) including Hamilton [14], NeuroKit [27], Pan and Tompkins [13] and Pan-Tompkins++ [26] for an ECG segment and (b) ECG segment of the subject 2 of GUDB during Maths Test, recorded using chest strap with V2-V1.

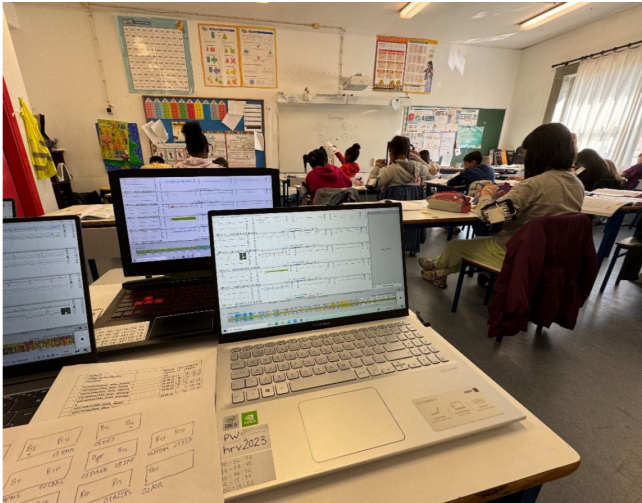


Fig. 3. RegularMente data acquisition setup using BITalino by PLUX and OpenSignals software [30].

TABLE V
EVALUATION RESULTS FOR THE HRV PARAMETERS EXTRACTED WITH ADAPTIVE ACC FILTERING.

Time and Frequency Domain Evaluation Metrics with Adaptive ACC Filtering				
Parameter	Unit	Our Work	Kubios HRV	$ \Delta_x $
Mean HR	[bpm]	258.28 $\pm$ 56.89	109.52 $\pm$ 10.90	148.76
Minimum HR	[bpm]	50.42 $\pm$ 0.00	34.92 $\pm$ 6.83	15.50
Maximum HR	[bpm]	240.00 $\pm$ 0.00	240.00 $\pm$ 0.00	0.00
Mean NN Interval	[ms]	570.02 $\pm$ 42.40	551.94 $\pm$ 51.58	18.08
RMSSD	[ms]	222.28 $\pm$ 52.47	111.45 $\pm$ 38.44	110.83
pNN50	[%]	38.57 $\pm$ 15.33	31.02 $\pm$ 13.37	7.56
SDNN	[ms]	154.76 $\pm$ 35.42	117.99 $\pm$ 40.09	36.77
Low Frequency Power (LF)	[ms ²]	211370.44 $\pm$ 115065.49	9224.40 $\pm$ 9061.34	202146.04
High Frequency Power (HF)	[ms ²]	734424.73 $\pm$ 344728.96	6902.37 $\pm$ 6433.32	727522.36
Total Power	[ms ²]	422364.78 $\pm$ 231487.04	17241.92 $\pm$ 16225.69	405122.87
LF/HF Ratio	[—]	0.28 $\pm$ 0.03	1.45 $\pm$ 0.56	1.17
TINN	[ms]	170.31 $\pm$ 117.50	1349.60 $\pm$ 273.89	1179.29

TABLE VI
EVALUATION RESULTS FOR THE HRV PARAMETERS EXTRACTED WITH NOISE DETECTION.

Time and Frequency Domain Evaluation Metrics with Noise Detection				
Parameter	Unit	Our Work	Kubios HRV	$ \Delta_x $
Mean HR	[bpm]	115.15 $\pm$ 14.76	109.52 $\pm$ 10.90	5.63
Minimum HR	[bpm]	55.38 $\pm$ 6.80	34.92 $\pm$ 6.83	20.47
Maximum HR	[bpm]	210.07 $\pm$ 43.92	240.00 $\pm$ 0.00	29.93
Mean NN Interval	[ms]	597.04 $\pm$ 61.40	551.94 $\pm$ 51.58	45.09
RMSSD	[ms]	86.07 $\pm$ 53.02	111.45 $\pm$ 38.44	25.37
pNN50	[%]	25.26 $\pm$ 14.53	31.02 $\pm$ 13.37	5.76
SDNN	[ms]	81.31 $\pm$ 37.31	117.99 $\pm$ 40.09	36.68
Low Frequency Power (LF)	[ms ²]	19501.06 $\pm$ 22219.73	9224.40 $\pm$ 9061.34	10276.66
High Frequency Power (HF)	[ms ²]	89024.45 $\pm$ 94482.91	6902.37 $\pm$ 6433.32	82122.08
Total Power	[ms ²]	56807.10 $\pm$ 64819.90	17241.92 $\pm$ 16225.69	39565.19
LF/HF Ratio	[—]	1.70 $\pm$ 1.84	1.45 $\pm$ 0.56	0.24
TINN	[ms]	155.87 $\pm$ 47.35	1349.60 $\pm$ 273.89	1193.73